

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of MSc Physics Examination March 2018

Course code & Name PS905 Atomic Physics

Date: 27-03-2108 Day: Tuesday Time: 1.30 p.m. To 1.50 p.m. Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Que.1 Tick the correct answer.

- (a) The hydrogen atom is in p-state. For this the value of j are
1. $5/2, 3/2, 1/2$
 2. $3/2, 1/2$
 3. $-1/2, 1/2, 3/2$
 4. $-1/2, -3/2, -1/2$
- (b) The parity of ${}_8\text{O}^{16}$ is
1. Odd
 2. Even
 3. None of the above
- (c) For $J = 5/2$ how many degenerate states are possible.
1. 2
 2. 6
 3. 8
 4. 10

- (d) Which of the following perturbation increase with Z^2
1. Residual electrostatic interaction
 2. Spin-orbit interaction
 3. Effect of other external fields or effect of nucleus
 4. Central field potential
- (e) When the LS coupling breaks down in Mg, which of the following transitions occurs.
1. Ultraviolet transition between triplet and triplet states.
 2. Optical transitions between singlet and triplet states.
 3. Both (1) and (2).
 4. None of the above.
- (f) Splitting of atomic energy levels in a weak magnetic field is known as,
1. Paschen-Back effect
 2. Stark effect
 3. Zeeman effect
 4. Photo-electric effect.
- (g) Classify the following broadening in terms of homogeneous and nonhomogeneous categories.
1. Doppler broadening
 2. Natural broadening
 3. Pressure broadening
 4. Crystal field broadening
- (h) Transition from 1F_3 to 1D_2 under magnetic field gives:
1. Normal Zeeman effect
 2. Anomalous Zeeman Effect.
 3. None of the above
- (i) Spin Multiplicity of $^2D_{3/2}$ is
1. 2
 2. 1
 3. 3
 4. 4
- (j) 3p state follows the interval rule: True/False

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III Semester of MSc Physics Examination March 2018

Course code & Name PS905 Atomic Physics

Date: 27-03-2108 Day: Tuesday Time: 1.50 p.m. To 3.30 p.m. Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
 2. Make suitable assumptions and draw neat figures wherever required.
 3. Use of non-programmable calculator is allowed.
 4. Show necessary calculations.
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Section-I

- Que.2** (a) Calculate the first four term values for Hydrogen atom in terms of eV. The R_∞ is $1.09737 \times 10^{-7} \text{ m}^{-1}$ and ground state term value for hydrogen is -13.6 eV. (2)
- (b) Give electronic configuration for nl (n is a principal quantum number and l is orbital quantum number) value of the last electron in the subshell for an atom of $_{10}\text{Ne}$ and $_{17}\text{Cl}$. (3)
- (c) What are the singlet and triplet state? Explain in brief with example. (2)
- (d) Explain in brief Normal Zeeman effect and how it differ from anomalous Zeeman Effect. (3)

Section-II

- Que.3** (a) What is Parity? Prove that Eigen value of parity operator is ± 1 . (4)
- (b) Calculate the $\langle r \rangle$ in the ground state of hydrogen atom. (4)
- (c) Calculate the angle between L and S for $^2D_{5/2}$ term. (3)
- (d) Draw the energy level splitting of level $3s3p^1P_1$ by a transition to $3s3d^1D_2$ under the influence of weak magnetic field and show allowed transitions. (4)
